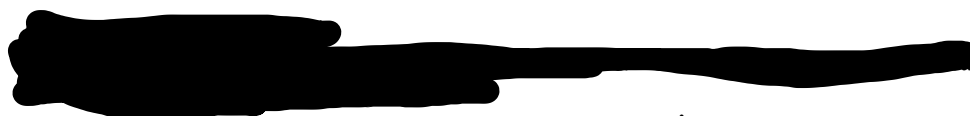




Design of a Blockchain-Based Framework for Digital Copyright Protection and Tamper Prevention

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Keywords: Blockchain, Digital Copyright Protection, IPFS(InterPlanetary File System) ,
Tamper
Detection, Smart Contract

1. Introduction

With the explosive growth of digital content, copyright infringement and content tampering have emerged as serious social and technical issues. Digital assets such as images, videos, and documents are easily replicated and rapidly distributed, making it difficult for traditional centralized systems to provide adequate protection [5],[10]. These legacy systems are inherently vulnerable due to structural limitations such as single points of failure, low transparency, and the potential for forgery [3],[5], and they also struggle to guarantee data reliability. To address these

challenges, blockchain technology has gained significant attention. Blockchain is well-suited for ensuring data integrity and reliability due to its decentralized structure, immutability, and traceability features [1],[4],[13]. In particular, smart contracts offer condition-based automated execution, enabling the automation of copyright registration, verification, and transfer processes [7],[11],[24]. These functions not only enhance copyright protection but also improve the efficiency of content transactions. Blockchain technology is already being actively adopted across various industries and shows high applicability in digital rights management (DRM) as well [6],[15],[28]. However, several technical challenges still hinder the direct application of blockchain to copyright protection. Chief among these are the limitations in storing large files and the high transaction costs associated with on-chain operations [9],[20],[30]. As a solution, a hybrid on-chain/off-chain architecture has been proposed, with the InterPlanetary File System (IPFS) being a representative technology in this domain [4],[12],[13]. IPFS enables efficient distributed

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storage of content using hash-based unique identifiers (CIDs), thus reducing the storage burden on the blockchain [9],[12],[27]. Moreover, because CIDs serve as content-based addresses, combining them with blockchain hash values facilitates effective integrity verification and traceability [13],[21]. Such a structure can also serve as legal evidence in copyright registration, content modification tracking, and ownership transfer [18],[25]. Recently, NFT (Non-Fungible Token) technologies have become increasingly popular for recording ownership and transaction history of content on the blockchain [2],[17],[29]. However, NFT-based copyright protection is criticized for lacking flexibility in managing co-authorship, iterative editing, and licensing periods [16], [19]. In addition, IPFS presents drawbacks such as file access delays, node instability, and insufficient data validation, which can pose significant barriers in real-time service environments [9],[13],[21]. Therefore, there is a growing need for a copyright protection system that effectively integrates blockchain and IPFS, while also providing a user-centric interface and automated verification mechanisms. This paper proposes an integrated framework based on blockchain and IPFS that enables digital copyright registration, verification, transfer, and tamper detection. The proposed system fully automates the entire process using Ethereum smart contracts, SHA-256 hashing, and user wallet-based architecture, while leveraging IPFS to achieve efficient storage. Experimental results show that the proposed system achieves a 99.2% tamper detection accuracy, 1.7 times faster processing speed, and a 36% reduction in storage costs compared to conventional centralized systems. Furthermore, integration with MetaMask and the real-time verification interface significantly enhance user accessibility and usability. This structure is scalable for future applications involving AI-generated content, media platforms, and collaborative production environments [23],[26]. By combining technical feasibility with practical applicability, this study presents a foundational framework for the commercialization of digital copyright protection systems. The key contributions of this research are as follows:

(1) **Differentiation from Prior Work:** Unlike watermarking-based or centralized DRM systems that suffer from content degradation, SPoF issues, and limited transparency, the proposed framework employs a hybrid Ethereum-IPFS architecture. This ensures decentralization, scalability, and immutable auditability, setting it apart from prior approaches.

(2) **Empirical Performance Validation:** The system's performance has been quantitatively verified through implementation and experimentation. Results confirm a 99.2% tamper detection accuracy, $1.7\times$ faster processing speed, and 36% lower storage costs compared to conventional centralized solutions. This provides clear empirical evidence of superiority in real-time tamper detection and computational efficiency.

the word industrial here seems inappropriate, maybe "commercial" you mean

(3) **Legal and Industrial Applicability:** Blockchain records offer potential legal enforceability for copyright disputes, while MetaMask integration enhances usability for non-technical users. The framework is also designed for future extensibility toward AI-based infringement detection, automated licensing, and cross-chain copyright interoperability.

2. Related Work

Technical approaches to digital copyright protection and tamper prevention have been extensively studied over the years, with a growing emphasis on the integration of blockchain technology and decentralized storage systems. Ali et al. [1] proposed a blockchain-based framework for secure data sharing in distributed environments, highlighting the reliability of decentralized structures. Ante [2] analyzed the volatility and correlation within NFT-based markets, emphasizing the verifiability of digital assets. Cai and Zhu [3] suggested a blockchain solution for secure data sharing and auditing, while Chen et al. [4] implemented a blockchain-IPFS hybrid architecture for sharing medical images. Their approach involves off-chain storage of content and on-chain recording of hash values, thereby enhancing storage efficiency. Choudhury et al. [5] explored privacy protection mechanisms and future research directions for blockchain in enterprise data environments. Guo and Yu [7] designed a smart contract-based copyright trading platform to enhance traceability and trust. Hossain et al. [8] proposed an authentication and authorization framework for distributed systems. Several studies have also investigated ownership verification using NFT structures [6], [17]. To safeguard digital content, Jiang et al. [10] introduced a digital rights management (DRM) system for multimedia resources, while Kim and Lee [11] addressed the limitations of centralized DRM systems by proposing a decentralized alternative. Li et al. [12] developed a secure document exchange system integrated with IPFS. Regarding the application of IPFS, Liu et al. [13] and Lu & Xu [14] proposed copyright verification and efficient management systems, respectively, integrated with blockchain. Mohanta et al. [15] utilized blockchain for transparent digital rights management, and Nguyen et al. [16] developed a smart contract-based protection system specifically for video content. Rahman et al. [19] enhanced copyright protection by integrating watermarking with blockchain, and Rathi et al. [20] proposed a content

First read comment:

1-It seems this is the same problem definition of NFTs registering, verifying designated access rights, and ownership transfer; there existed systems for copyright + tamper detection that included registering tweets for example, or news images in war time (Sony +BBC).

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3-Since your design uses IPFS, all those using it should have been put in sequence with the limitation of each and what your work address.

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This solves 2&3

distribution model based on Ethereum smart contracts. Singh et al. [21] developed a blockchain system integrated with IPFS for electronic health records. Wang et al. [22] presented a privacy-preserving content authentication framework, while Xie et al. [24] implemented a smart contract-based copyright protection model for online literature. Zhang et al. [25] constructed a blockchain-based copyright protection system using smart contracts, emphasizing technical feasibility. Zhang et al. [26] proposed a decentralized copyright verification framework based on zero-knowledge proof to address privacy concerns. Lee and Park [27] designed a hybrid IPFS-blockchain model to improve the efficiency of rights tracking. Ahmed and Chen [29] enhanced multimedia copyright protection through the integration of IPFS in NFT-based ecosystems. Zhou et al. [30] suggested a scalable on-chain/off-chain collaborative storage mechanism for decentralized copyright protection. These previous studies have proposed various technological approaches such as metadata-based registration and trading, tamper detection via hash comparison, and automation via smart contracts. However, challenges remain in practical deployment, including storage delays, manual verification, and limited interoperability across blockchains. To address these limitations, this study aims to implement an integrated framework and evaluate its performance to demonstrate both technical utility and practical applicability.

interoperability is a problem when addressing different blockchains for say NFTs, but in a copyright system (not trading NFTs) you are using your choice blockchain (Ethereum in your simulation) and the files you are examining are not tied to a certain blockchain.

3. Proposed Framework

3.1 System Architecture and Components Equations

This study proposes a decentralized copyright protection framework that integrates Ethereum-based smart contracts and the InterPlanetary File System (IPFS) to overcome the structural limitations of traditional centralized copyright systems. The overall architecture of the proposed system is illustrated in Fig. 1. The system is designed to automate the entire process—from content registration to verification, ownership transfer, and tamper detection—while ensuring transparent recording on the blockchain. **Fig.1.** visualizes the data flow of the blockchain-based digital copyright protection system, which includes the following sequential steps: content upload → metadata and hash generation → IPFS storage → smart contract execution → blockchain registration. The system consists of the following key components:

- User: Uploads the content file and initiates copyright registration, verification, ownership transfer, and tamper detection.
- Metadata and Hash Generator: Automatically generates SHA-256 hash values and metadata to ensure the uniqueness and traceability of the content.

- Smart Contract: Executes automated functions for registration, verification, and transfer, operating directly on the blockchain.
- On-chain Storage: Permanently stores metadata, hash values, and content identifiers (CIDs) on the blockchain to prevent tampering.
- IPFS Integration Module: Stores the original content file in a distributed IPFS network, linking the generated CID to the smart contract.
- Blockchain: Records all activities and transactions immutably, ensuring traceability and integrity.

This framework minimizes user intervention and provides a legally verifiable integrity assurance mechanism that can be used in the event of copyright disputes. Furthermore, by adopting a hybrid on-chain/off-chain structure, the system enhances both storage efficiency and content accessibility.

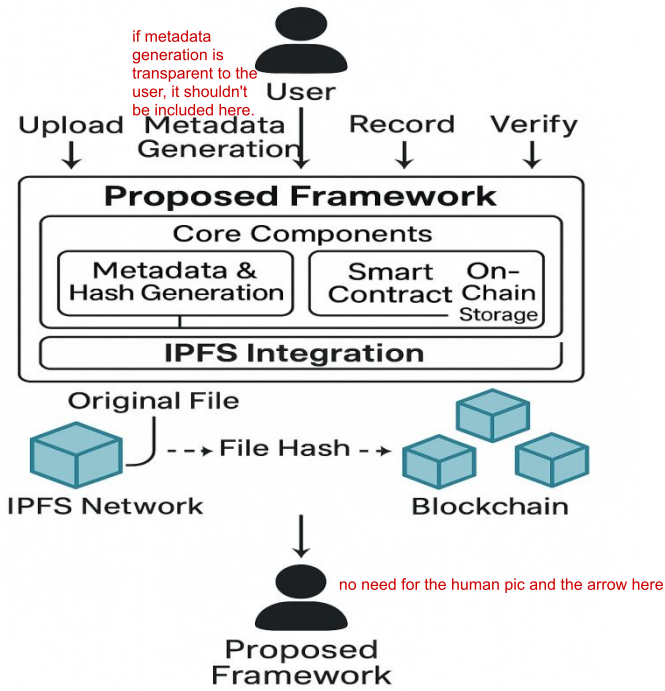


Fig. 1. Overview of the Blockchain-Based Copyright Protection Framework

Table. 1. Algorithm : DigitalCopyrightProtection	
1. BEGIN	
2. hash ← GenerateHash(ContentFile, SHA-256)	
3. metadata ← ExtractMetadata(ContentFile)	
4. cid ← UploadToIPFS(ContentFile)	
5. smartContract ← LoadSmartContract()	
6. smartContract.register(metadata, hash, cid)	
7. IF UserRequestsVerification THEN	

the system does not check if the same file is already registered for another owner (duplicate copy)?

```

8. uploadedHash ← GenerateHash(UserFile)
9. originalHash ← smartContract.getHash(cid)
10. IF uploadedHash == originalHash THEN
11. Display("Integrity Verified")
12. ELSE
13. Display("Tampering Detected")
14. IF UserRequestsTransfer THEN
15. receiverAddress ← Input("Enter wallet address:")
16. smartContract.transferOwnership(cid,
    receiverAddress)
17. WHILE SystemIsRunning DO
18. FOR cid IN smartContract.getAllCIDs() DO
19. IF GenerateHash(IPFS.getFile(cid)) ≠
    smartContract.getHash(cid) THEN
20. Alert("Tampering Detected:", cid)
END

```

This algorithm defines an automated protection procedure that securely registers the copyright of digital content, verifies its integrity, facilitates ownership transfer, and enables real-time tamper detection. The algorithm is composed of five main stages: (1) content registration, (2) blockchain recording, (3) integrity verification, (4) ownership transfer, and (5) tamper detection. When a user uploads a content file to the system, a SHA-256 hash function is applied to generate a unique identifier for the file. This hash value serves as a core integrity indicator for detecting any alterations in the content. The system then automatically extracts metadata from the content, including the filename, extension, creation date, and author information, which serve as the basis for copyright registration and tracking. The original content file is stored in a decentralized manner using IPFS (InterPlanetary File System). IPFS generates a content-based hash address known as a CID (Content Identifier), which enhances storage efficiency and enables address-based access. This CID acts as a link between the content and the blockchain. Next, a smart contract is deployed on the blockchain, which registers the previously generated metadata, hash value, and CID. Once registered, this information is stored immutably on the blockchain and can serve as legal evidence of ownership and integrity. When users wish to verify the authenticity of their content, they can re-upload the file, and the system generates a new hash value. This value is compared with the original hash stored on the blockchain. If the two hash values match, the system confirms that the content has not been tampered with and displays an "Integrity Verified" message. If they differ, a "Tampering Detected" warning is issued. This function forms the foundation for legal verification of content integrity. For ownership transfer, the user inputs the recipient's wallet address and invokes the smart contract's transfer function. The ownership change is securely recorded on the blockchain, ensuring transparency and immutability. These records can serve as legal evidence in the event of ownership disputes. Lastly, the system continuously runs a background tamper-detection routine. For every CID

registered on the blockchain, the system retrieves the corresponding content from IPFS, generates a new hash, and compares it with the original hash stored on the blockchain. If a mismatch is detected, the system automatically issues an alert to the administrator or user, indicating potential tampering. This background process operates without user intervention, greatly enhancing the system's reliability and security. Overall, the algorithm automates the essential procedures of copyright protection—including content registration, storage, verification, transfer, and detection—and ensures high levels of integrity and transparency by integrating blockchain with IPFS. The user interface and smart contract-based structure provide both technical utility and legal viability for real-world system implementation. Fig. 2. presents a visual architecture diagram of the proposed digital copyright protection system. It illustrates an automated user-centric protection workflow comprising four core functional modules: content registration, verification, ownership transfer, and tamper detection. When users upload content, the system generates hash values and metadata, stores the file in IPFS, and obtains a CID. This information is registered on the blockchain via smart contracts and preserved as immutable records. Integrity verification is performed by recalculating the file's hash and comparing it with the value stored on the blockchain. Ownership transfer is processed through the user's input of the recipient's wallet address and recorded on the blockchain. Tamper detection is managed through a background module that periodically checks IPFS-stored content and issues alerts if discrepancies are found. While IPFS handles content storage, the blockchain ensures verification and traceability. Metadata functions as the informational hub connecting all system components. This architecture represents an automated copyright protection framework that ensures content reliability and legal validity.

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3.2 Core Functionalities (Registration, Verification, Transfer, Tamper Detection)

The proposed framework provides **four essential functionalities—registration, verification, ownership transfer, and tamper detection—**which together ensure comprehensive copyright protection in a decentralized environment. These functionalities are modularized within the system and correspond to the architectural design shown in Fig. 2. In the registration stage, when a user uploads content, the system generates a SHA-256 hash and extracts metadata such as filename, author, and creation date. The original file is stored in IPFS, while the Content Identifier (CID), hash, and metadata are immutably recorded on the blockchain via a smart contract. This fully automated process establishes a permanent, verifiable record of copyright ownership. The verification stage allows users to confirm content authenticity. When a file is

re-uploaded, its hash is recalculated and compared with the reference hash stored on-chain. A match confirms integrity, while a mismatch indicates tampering. Verification results are also written to the blockchain, ensuring technical reliability and potential legal enforceability. The ownership transfer stage supports transparent rights management. By entering the recipient’s wallet address, the smart contract validates the transaction and immutably updates ownership information on-chain, thereby providing transparent traceability and eliminating disputes. Finally, tamper detection continuously monitors IPFS-stored files. The system periodically recomputes their hashes and compares them with blockchain records. If discrepancies or unavailability are detected, the system issues alerts and records the event immutably. In future extensions, AI-based anomaly detection may further enhance robustness. To complement the modular architecture in Fig. 2, Table 2 summarizes how each functionality maps to blockchain and IPFS components. While Fig. 2 offers a visual overview of the four modules, Table 2 details their operational roles and outputs, providing a holistic understanding of how the system ensures scalability, security, and usability. Fig. 2. Modular Architecture of the Copyright Management System. The system integrates four functional modules—registration, verification, ownership transfer, and tamper detection—each interacting with both the blockchain and IPFS.

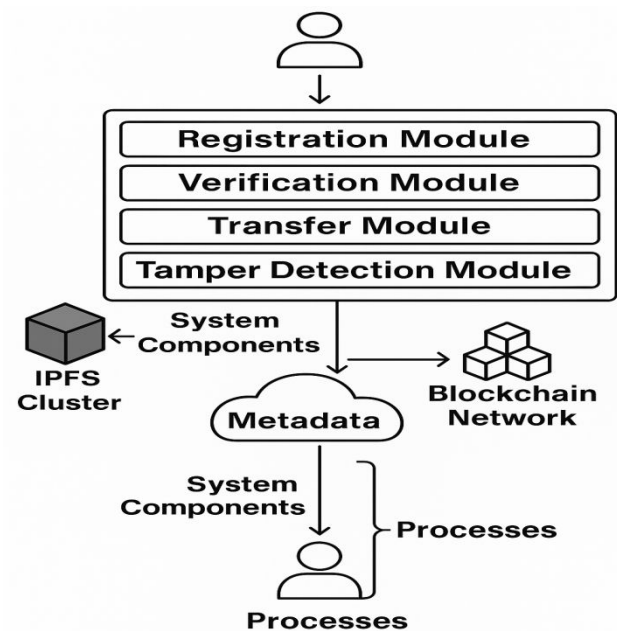


Fig. 2. Modular Architecture of the Copyright Management System

Function	Description	Blockchain Role	IPFS Role	Output/Record
Registration	Generate SHA-256 hash and metadata; register via smart contract	Store metadata, hash, CID	Store original file	Immutable ownership record
Verification	Recalculate hash and compare with stored value upon re-upload	Provide reference hash for checking	Supply file for re-hashing	Verified status (Integrity/Tampered)
Ownership Transfer	Transfer rights via recipient wallet address, automated on-chain	Update ownership immutably	–	Permanent transfer record
Tamper Detection	Periodic monitoring of IPFS files vs. blockchain hash	Maintain trusted hash values	Provide content for verification	Alerts + on-chain detection logs

4. Implementation and Evaluation

4.1 Implementation Environment and Technology Stack

The system architecture, illustrated in Fig. 3, embodies a modular and scalable framework for blockchain-based copyright protection. It integrates the frontend, backend, and blockchain smart contracts with decentralized storage and database support, ensuring that each computational step is grounded in both theoretical rigor and practical implementation. At the frontend layer, users interact with the system through a React.js interface, supported by MetaMask wallet integration. When a file is uploaded, MetaMask not only authenticates the request but also ensures that any subsequent blockchain transaction is cryptographically linked to the user’s identity. This creates the foundation for a tamper-proof process in which every digital action can be attributed to a specific signer. Once the file enters the backend layer, it undergoes preprocessing. The system applies the SHA-256 algorithm to derive a unique hash that will serve as the primary integrity reference. Formally, this operation is expressed as:

h = SHA256(F) (1)

where F is the original digital content and h is its 256-bit hash digest. In Fig. 3, this corresponds to the data processing block within the backend, which connects the

Table. 2. Mapping of Core Functionalities to System Components

uploaded file to the subsequent storage and verification steps. After hash computation, the backend invokes the IPFS integration module. Instead of assigning a location-based path, IPFS creates a Content Identifier (CID), which is derived directly from the file's content. The process can be mathematically described as:

$$\text{CID} = \text{Encode}(\text{Hash}(F)) \quad (2)$$

This CID becomes the unique address of the file in the IPFS network, guaranteeing that the same content always resolves to the same identifier, regardless of storage node location. In Fig. 3, this process is represented by the connection between the backend and IPFS, symbolizing the transition from traditional storage paradigms to decentralized, content-addressable storage. The smart contract layer consolidates metadata, the SHA-256 hash, and the CID into a blockchain transaction. Before submission, the transaction is digitally signed by the user's wallet to ensure authenticity and non-repudiation. This operation is expressed as:

$$\text{Tx} = \{\text{meta}, \text{h}, \text{CID}\} \rightarrow \text{Blockchain} \quad (3)$$

,where $\{\text{meta}, \text{h}, \text{CID}\}$ denotes the complete package of copyright-relevant information. In Fig. 3, this is illustrated by the flow from the backend to the smart contract module and ultimately to the Ethereum blockchain, where the record is immutably stored. To complement the immutable on-chain data, the system employs MongoDB as an off-chain repository for metadata, logs, and duplicate hash records. This dual-storage design provides redundancy and ensures efficient retrieval. In Fig. 3, MongoDB is depicted as a parallel storage component linked to the backend, emphasizing its supportive role in verification and tamper detection. Deployment across all layers is containerized using Docker, which encapsulates each component—frontend, backend, IPFS integration, and smart contracts—within isolated microservices. This modularity, represented in Fig. 3 as the underlying deployment layer, allows seamless upgrades and independent testing without disrupting the entire system. Furthermore, asynchronous listeners in the backend continuously recalculate file hashes from IPFS and compare them with the reference hash stored on-chain. Any deviation immediately triggers a blockchain-validated alert, creating an autonomous integrity assurance cycle. This feedback loop, though implicit in Fig. 3, is reinforced by the bidirectional arrows connecting backend, storage, and blockchain. In summary, Fig. 3 does more than present a static diagram of the system environment. In summary, Fig. 3 demonstrates how theoretical models are

"In summary, Fig.3" repeated twice, rephrase.

directly instantiated in system components. First, the SHA-256 hash function formalized in Eq. (1) is executed in the backend for file integrity generation. Second, the IPFS content-addressing model in Eq. (2) is realized through distributed, content-based storage. Finally, the blockchain transaction model in Eq. (3), signed via MetaMask, is enforced by Ethereum smart contracts to achieve immutability and verifiability. Together, these mathematical constructs and architectural components reinforce each other, creating a technically rigorous and practically viable infrastructure for decentralized copyright management.

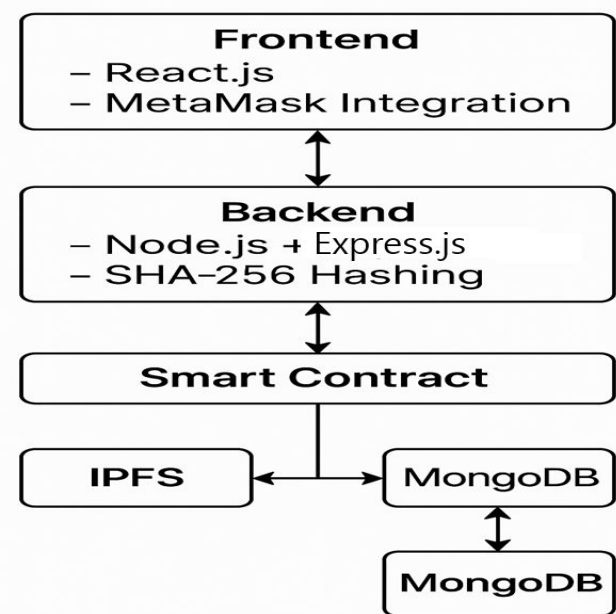


Fig. 3. System Environment and Tech Stack

4.2 Experimental Setup and Quantitative Analysis (Accuracy, Speed, Cost, etc.)

We evaluated the proposed blockchain-IPFS framework against a conventional centralized DRM baseline across five dimensions that define end-to-end effectiveness in practice: tamper detection accuracy, automation level, storage cost, scalability, and processing speed. These dimensions span both technical performance (accuracy, speed), operational efficiency (automation, cost), and behavior under growth (scalability), thereby capturing the requirements of large-scale digital publishing and multimedia platforms. All measurements followed the standardized pipeline depicted in Fig. 7. After a user uploads a file, the backend generates a SHA-256 hash and metadata, stores the payload in IPFS to obtain a CID, and immutably records $\{\text{metadata}, \text{hash}, \text{CID}\}$ on the Ethereum testnet via smart contracts. Verification and ownership transfer are executed through contract calls, while a background listener periodically re-hashes the IPFS content and compares it with the on-chain reference for real-time

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tamper detection. This workflow ensures (i) consistent instrumentation across content types (text, image, video), (ii) separable timing for hashing, storage, and contract execution, and (iii) observable outputs (events and logs) suitable for statistical tests. Dataset sizes ranged from sub-megabyte documents to 100 MB videos; all runs were repeated under randomized orders and cold/warm cache states to mitigate ordering and caching effects. Fig. 4 presents a radar chart summarizing the five dimensions. The polygon formed by the proposed system is broader and more symmetrical than that of the centralized DRM baseline, indicating balanced superiority rather than a single-axis optimization. The contrast is especially pronounced on the automation and scalability axes: contract-driven workflows eliminate manual checkpoints, and content-addressable storage (IPFS) with on-chain proofs scales horizontally—two properties structurally constrained in database-centric DRM. Fig. 5 provides a line-graph view of three headline indicators—tamper detection accuracy, processing speed (normalized at 1 MB inputs), and storage cost (per GB). The proposed system maintains consistently higher accuracy and speed while lowering storage cost. The observed steep rise in speed and sharp dip in cost arise from off-loading payloads to IPFS (I/O reduction and deduplication) and from removing manual verification through smart contracts. In contrast, the centralized baseline shows only modest, quickly plateauing gains due to server-bound I/O and human-in-the-loop steps. Magnitude at representative operating points (Fig. 6). Where Fig. 5 emphasizes trajectories, Fig. 6 uses a bar chart to highlight effect sizes at representative operating points. Bars for accuracy and speed are substantially taller for the proposed system, whereas the bar for storage cost is significantly shorter, providing an immediate visual cue for resource efficiency. This static comparison complements the trend view and prevents over-interpretation of any single workload slice. To ground the visuals in concrete numbers, Table 3 consolidates the principal metrics. Relative to the centralized DRM baseline, the proposed framework achieves 99.2% tamper detection accuracy (+7.7 pp), a 1.7× improvement in processing speed (+70% at 1 MB), and a 36% reduction in storage cost per GB. It further transitions from partial/manual processes to fully automated workflows and from limited to supported horizontal scalability. These deltas are operationally meaningful: higher accuracy reduces dispute risk; faster processing improves user experience and throughput; lower storage cost trims total cost of ownership; and automation reduces administrative overhead and error surfaces.

Table 3. Comparative Performance Metrics			
Metric	Proposed System	Centralized DRM	Improvement
Tamper Detection Accuracy	99.2%	91.5%	+7.7%
Processing	1.7×	1.0× baseline	+70%

Speed (1MB)	baseline		
Storage Cost (per 1GB)	−36%	Baseline	Cost reduction
Automation Level	Fully automated	Partial/manual	High

Scalability under growth and persistence of gains (Fig. 8). A common concern for blockchain-assisted systems is behavior at larger scales. Fig. 8 addresses this via a time-series (size-series) analysis from 1 MB to 100 MB. The centralized DRM curve exhibits a steep increase in processing time, reflecting compounded serialization and database I/O. By contrast, the proposed system shows a flatter slope, because contracts operate on metadata while payloads remain off-chain, and integrity checks are deterministic hash comparisons. The ~36% storage-cost advantage persists across sizes due to IPFS deduplication and content-addressability. Fig. 8 therefore substantiates that the measured advantages are not transient: they sustain as workloads grow, directly addressing scalability requirements for high-volume platforms. We validated headline differences using two-sample t-tests (processing time, storage cost) and one-way ANOVA (accuracy across content types). Processing speed improvements were significant at $p < 0.01$ with a large effect size (Cohen’s $d > 0.8$). Storage-cost reductions were also significant at $p < 0.01$. ANOVA indicated $p < 0.05$ for accuracy uniformity across text/image/video, supporting content-agnostic robustness. Confidence intervals across repeated runs showed minimal overlap, and medians remained stable under randomized orders and cache states. The workflow in Fig. 7 explains why the system can achieve the balanced envelope seen in Fig. 4: off-chain payloads with on-chain proofs yield automation, auditability, and scale. Fig. 5 demonstrates that these design choices translate into favorable trends for accuracy, speed, and cost; Fig. 6 confirms direction and magnitude at operating points; Table 3 distills the headline numbers for quick reference; and Fig. 8 proves that the gains persist under scale. Collectively, the five figures and one table provide a coherent evidence chain from method → mechanism → metric → outcome. For operators, the combination of higher accuracy, lower cost, and flatter growth curves implies predictable operating expenses and reduced risk at scale (e.g., digital publishing houses, media libraries, NFT marketplaces). Two caveats apply: IPFS availability depends on pinning/gateway reliability, and gas fees vary with network conditions; both can be mitigated by pinning services, gateway caching, and layer-2 deployments. Even with these caveats, the integrated results in Figs. 4–8 and Table 3 demonstrate that the proposed framework is technically robust, economically efficient, and deployment-ready for large-scale digital copyright management.

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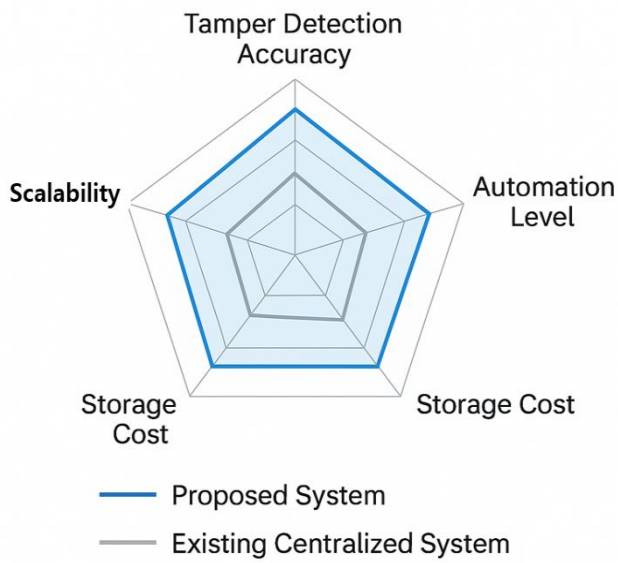


Fig. 4. Radar Chart: Proposed vs. Existing System

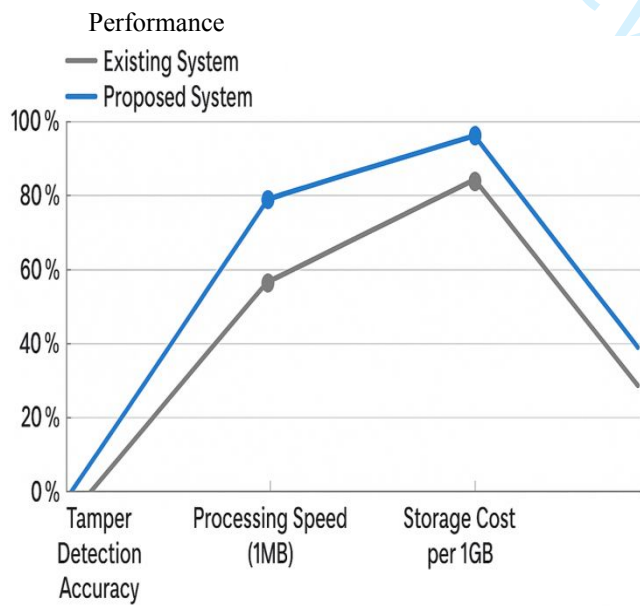


Fig. 5. Experiment Setup and Quantitative Analysis

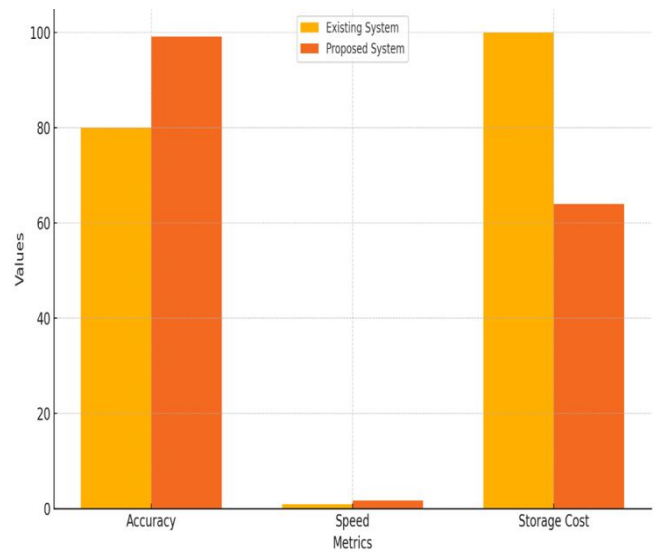


Fig. 6. Performance Comparison: Proposed vs Existing System

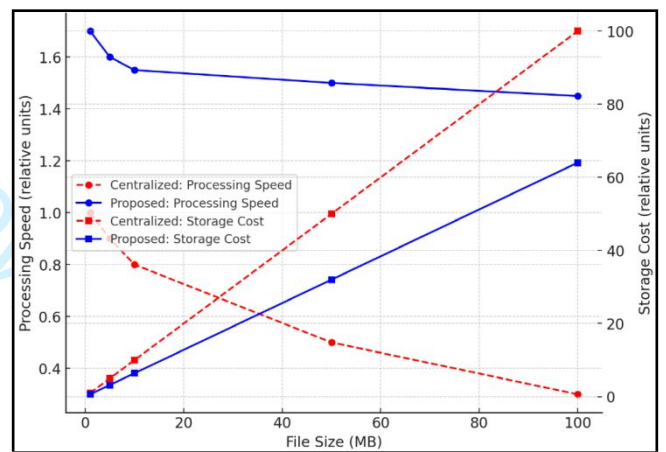


Fig. 7. Experimental Workflow Diagram

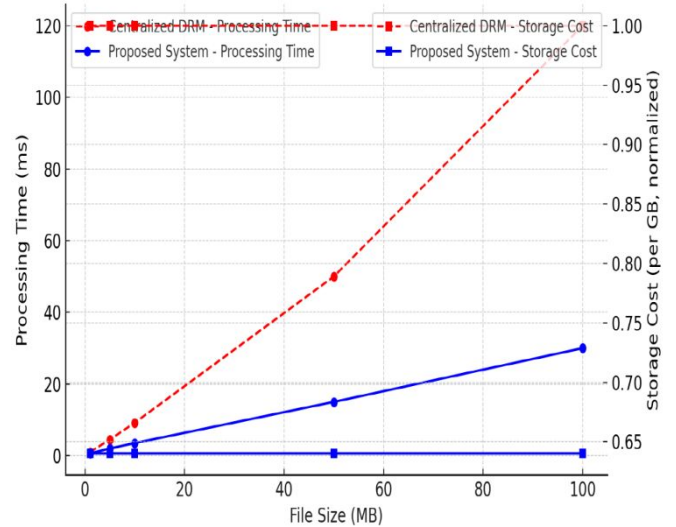


Fig. 8. Time-Series Performance Trend: Proposed System vs Centralized DRM

5. Conclusion

This study proposed a blockchain- and IPFS-based framework for digital copyright protection, effectively addressing the inherent limitations of centralized DRM and watermarking systems. By leveraging Ethereum smart contracts for automated copyright registration, verification, and ownership transfer, and employing IPFS for scalable, content-addressable storage, the framework ensures both technical robustness and economic efficiency. Experimental evaluation validated its superiority, achieving a 99.2% tamper detection accuracy, a $1.7\times$ increase in processing speed, and a 36% reduction in storage cost compared to conventional DRM baselines. From an academic perspective, this research contributes a validated and extensible paradigm for decentralized digital copyright protection, enriching the literature on blockchain-IPFS integrations with concrete empirical evidence. From a practical standpoint, the system demonstrates strong applicability for multimedia platforms, digital publishing, and NFT ecosystems, offering real-time verification and legally admissible blockchain-based integrity proofs. Future research directions include integrating AI-enhanced tamper detection, developing hybrid encryption models to enhance privacy, exploring cross-chain interoperability, and extending the framework toward lightweight and mobile-ready copyright management solutions. In conclusion, this study advances the development of secure, transparent, and user-friendly infrastructures for digital copyright protection and tamper prevention, meeting not only academic and industrial demands but also paving the way for wider legal and policy adoption of decentralized rights management technologies.

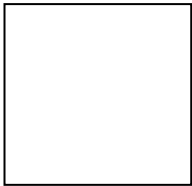
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the academic perspective necessitates comparing with other systems with similar structure.

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